

The Effect of Unconventional Oil and Gas Drilling on Healthcare Utilization: Evidence from Texas

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Abstract

This paper examines the relationship between inpatient healthcare utilization and the entry of unconventional drilling in Texas’s Eagle Ford region. Using a staggered difference-in-differences design, I find that inpatient claims for dermatological conditions, perinatal conditions, and pregnancy complications increase following ZCTA-level entry of unconventional drilling relative to not-yet-treated and never-treated ZCTAs. I then construct a novel dataset that combines public groundwater-source information with healthcare claims and well-location data to identify the effect of unconventional drilling near public drinking-water infrastructure. I find that communities with unconventional drilling near a public groundwater source experience greater increases in inpatient claims compared to communities with unconventional drilling farther away, including an aggregate increase of approximately 168 claims per ZCTA-quarter at the five-kilometer threshold. Perinatal conditions and pregnancy complications follow a more localized spatial pattern, with larger estimates at shorter distances from public groundwater sources. These findings suggest that the health consequences of unconventional drilling depend not only on whether drilling occurs, but also on where it is located relative to population-serving infrastructure.

JEL Codes— Q30, Q53, I12, H23

Keywords— Fracking, Unconventional Drilling, Drinking Water, Groundwater, Social Determinants of Health

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1 Introduction

Hydraulic fracturing and horizontal drilling¹ transformed natural resource extraction. These technologies allowed oil and natural gas producers to reach resources in previously inaccessible shale formations. Over the past few decades, these methods significantly increased and expanded oil and natural gas production in the United States. However, the use of unconventional methods in resource extraction is not without controversy; supporters tout the economic benefits from increased employment and higher wages, while opponents question the environmental and health hazards resulting from these methods.

Hausman and Kellogg (2015) conduct a welfare analysis to measure the economic benefits of the hydraulic fracturing boom. The authors find significant increases in consumer surplus associated with lower natural gas prices, with an estimated \$48 billion in increased total surplus. However, the authors note that their welfare calculations exclude negative externalities, such as environmental damage and negative health effects, due to a lack of research. Although numerous studies identify potential pathways that could lead to poor health outcomes associated with these new technologies (Colborn et al. 2011; Epstein 2017; McKenzie et al. 2012; Shonkoff, Hays, and Finkel 2014), the research causally identifying negative health consequences resulting from unconventional oil and gas drilling is limited. A complete assessment of the welfare effects of unconventional drilling would consider health effects. However, hydraulic fracturing is still a relatively new technology and any health-related ramifications may take time to develop. Continually updating research on the health effects of hydraulic fracturing is essential for understanding its full impact.

In this paper, I combine inpatient healthcare claims data with unconventional well locations to identify the effect of unconventional drilling on healthcare utilization. Prior work shows that unconventional drilling negatively affects infants whose mothers lived near drilling sites during pregnancy (Currie, Greenstone, and Meckel 2017; Hill and Ma 2022; Hill 2024). I extend this work by estimating the effect of unconventional drilling on inpatient healthcare utilization using changes in exposure to hydraulic fracturing between 2007 and 2015 in Texas’s Eagle Ford region. Although the aggregate estimate is imprecise, I find significant increases in perinatal claims, dermatological claims, and claims related to pregnancy complications.

These effects could stem from exposure to unconventional drilling or from demographic and economic changes associated with shale development. For example, oil and gas workers may move to the region for employment, and shifts in population, income, or insurance coverage could bias the estimates upward. Therefore, I examine outcomes among women and among Medicare and Medicaid beneficiaries. This analysis yields positive estimates across all three groups, although the estimated increase is smaller for women. The directional consistency across populations less likely to experience the male-dominated migration associated

1. Hydraulic fracturing, particularly when combined with horizontal drilling, is commonly known as ‘fracking.’

with oil and gas development suggests that compositional changes alone do not explain the observed increases.

Next, I investigate the health effects of drilling near public drinking water infrastructure. I compare changes in healthcare utilization across ZCTAs located on the shale formation that rely on public groundwater sources and differ in their proximity to unconventional drilling. Specifically, I compare ZCTAs where qualifying drilling occurs within one to five kilometers of a public groundwater source with ZCTAs that also experience unconventional drilling, but whose public groundwater sources remain more than five kilometers from qualifying wells.

The results show that, among ZCTAs experiencing unconventional drilling, inpatient healthcare utilization increases more in communities where drilling occurs near public groundwater sources. At the five-kilometer level, I find significant increases in overall inpatient utilization and in several health categories including circulatory, respiratory, cancer, neurological, and gastrointestinal conditions. These estimates are generally similar in magnitude to those in the primary analysis but are more precisely estimated in the groundwater-source sample. Pregnancy complications and perinatal outcomes exhibit a more concentrated spatial pattern, with the largest pregnancy-complication estimate occurring when drilling takes place within one kilometer of a groundwater source and a significant increase in perinatal conditions when drilling occurs within three kilometers.

Because these estimates compare drilling near public groundwater sources with drilling located farther from those sources, the results isolate the additional health effect associated with drilling near drinking-water infrastructure among communities already experiencing shale development. The findings are therefore consistent with drilling near public groundwater sources generating additional local environmental exposures that increase healthcare utilization. The research design identifies the causal effect of drilling near public drinking-water sources relative to drilling farther from those sources, but it does not separately identify the environmental pathway responsible for that effect.

This paper adds to the growing literature on externalities related to unconventional drilling methods. One line of research studies the effect of natural gas development on property values and suggests that proximity to an unconventional drilling site is a disamenity. Muehlenbachs, Spiller, and Timmins (2015) find that the perceived risk of groundwater contamination negatively affects the property value of homes near gas wells, and particularly homes that depend on private groundwater wells for drinking water. The authors find that the property values of homes that use publicly supplied water are not affected or may even benefit from proximity to a producing well through royalty payments. Although the authors do not estimate groundwater contamination, their work suggests that individuals internalize the potential risk of contamination, driving down property values of affected homes. In a similar study, Balthrop and Hawley (2017) investigate home prices for homes largely dependent on public water, finding that houses closer (within

3,500 feet) to hydraulically fractured wells sell at lower prices than houses farther away. The authors suggest various reasons for the negative effect, such as pollution, risk of earthquakes, or visual disamenities, but they do not identify a mechanism. My results provide evidence that proximity of unconventional drilling to public water infrastructure has measurable health consequences, offering one potential explanation for why households may value distance from drilling activity.

A related literature documents environmental contamination associated with shale development, including both air and water pollution (Hildenbrand et al. 2016; Hill and Ma 2022; Litovitz et al. 2013; Osborn et al. 2011; Xu 2023). These studies establish the presence of pollutants near drilling activity, but provide more limited evidence on how these exposures translate into realized health outcomes for nearby populations. Much of the existing work in the economics literature focuses on maternal and infant health, finding that proximity to hydraulic fracturing is associated with adverse birth outcomes (Currie, Greenstone, and Meckel 2017; Hill and Ma 2022; Hill 2024). For example, Currie, Greenstone, and Meckel (2017) compare infant health outcomes before and after the entry of a hydraulically fractured well and find that mothers living within three kilometers of a well give birth to infants with worse health outcomes, including a 25 percent increase in the probability of low birth weight. Hill and Ma (2022) extend this work by showing that drilling near public water sources increases the risk of shorter gestation and low birth weight.

Epidemiological studies document associations between unconventional drilling and a wide range of adverse health outcomes. Black et al. (2021) synthesize this evidence, highlighting links between drilling activity and increased hospitalizations as well as disease incidence. Existing studies identify associations across multiple systems, including respiratory, dermatological, and genitourinary conditions, as well as more severe outcomes such as pediatric asthma hospitalizations and elevated cancer risk among children living near high concentrations of wells (Denham et al. 2019; Willis et al. 2018; McKenzie et al. 2017).

Despite this broad evidence of potential health effects, causal research remains much narrower in scope. Existing quasi-experimental studies are concentrated on infant and child health and on a small number of respiratory outcomes. As noted by Black et al. (2021), relatively few studies causally identify effects on adult health or examine the broader set of conditions that may respond to unconventional drilling. Peng, Meyerhoefer, and Chou (2018) find increases in pneumonia hospitalizations among elderly patients, while Xu (2023) documents increases in respiratory-related hospitalizations in Texas following drilling activity. This paper broadens the causal evidence by examining inpatient utilization across a broad set of health conditions and by testing whether these effects vary with the location of drilling relative to public drinking-water infrastructure.

I also extend prior research by examining healthcare utilization in an understudied region: the Eagle Ford Shale formation. Much of the existing causal evidence on the health effects of unconventional drilling comes

from the Marcellus Shale in Pennsylvania and New York.² Differences in geology across shale formations imply that drilling techniques and the composition of released pollutants vary by region (Speight 2019). In contrast to the Marcellus, which primarily produces natural gas, the Eagle Ford produces both oil and natural gas, leading to additional byproducts such as flaring-related emissions. These differences may alter both the type and intensity of environmental exposures.

In addition, the timing of development in the Eagle Ford provides useful variation for identification. Unlike the Marcellus, where portions of the resource were accessible through earlier conventional drilling, the Eagle Ford became economically viable only after its discovery in 2008, followed by a rapid expansion in drilling beginning around 2010. This rapid and staggered expansion in drilling provides substantial temporal and geographic variation in exposure, which I exploit using a staggered difference-in-differences design.

In the next section, I describe the history of the Eagle Ford Shale formation, the process of hydraulic fracturing and horizontal drilling, and potential pathways for negative health outcomes. In section three, I outline the data and provide descriptive statistics. Section four reviews my methods and results. Section five concludes.

2 Background

2.1 UNCONVENTIONAL DRILLING

Unconventional drilling refers to newer methods of extracting oil and natural gas from underground reservoirs that were inaccessible using previous technologies. While conventional drilling involves drilling vertically into oil or natural gas reservoirs, unconventional drilling uses alternative methods to release oil and natural gas trapped in tight geological formations. Two unconventional drilling methods, often used in combination, are hydraulic fracturing and horizontal drilling. The process of unconventional drilling involves several phases between preparation and production (Hill 2018). Once a drilling site is located and permitted, well pad construction begins, which includes clearing and preparing the area for drilling. Next, wells are drilled on the well pad, which can take up to two months for horizontally drilled wells. After drilling, hydraulic fracturing takes place for around seven days to stimulate oil and natural gas production. The hydraulic fracturing process involves injecting water, chemicals, and/or sand into rock formations to break up the rock and release the oil and gas. Conversations with oil and natural gas engineers suggest that after the vertical portion of the well reaches the target formation, the wellbore extends horizontally through the shale, and hydraulic fracturing is conducted along this lateral section of the well. After fracturing, there is a period of

2. One exception is Willis et al. (2021), which examines birth outcomes in Texas, though the health data end in 2009.

flowback. During the flowback period, injected fracturing fluid returns to the surface along with formation water and hydrocarbons. The resulting wastewater, including flowback and produced water, may be reused or transported for treatment or disposal in injection wells (Kondash, Albright, and Vengosh 2017).

Cost-effective hydraulic fracturing and horizontal drilling began in the late 1990s and early 2000s in the Barnett Shale region of Texas. By 2008, Texas produced more than 70 percent of shale gas in the United States (Gilmer and Kerr 2010). In addition to the Barnett Shale region, unconventional drilling technology opened new opportunities in several other Texas geological formations including the Permian Basin, the Haynesville Shale formation, and the Eagle Ford Shale formation. According to Gilmer, Hernandez, and Phillips (2012), drilling began in the Eagle Ford region in 2008 in La Salle County. Production subsided in 2009 as a result of declining oil and gas prices and the Great Recession, but picked up again in 2010 and grew rapidly, creating useful temporal and geographic variation to study the health consequences of unconventional oil and gas drilling. Though nearby formations were accessible using conventional techniques, oil and natural gas in the Eagle Ford Shale formation was not accessible prior to 2008, and therefore the new technology allowed for a sharp increase in production after discovery (Speight 2019). By 2021, Eagle Ford was one of the most productive formations in Texas. Figure 1 shows the increase in oil and gas wells drilled in the Eagle Ford formation following the expansion of horizontal drilling, and captures the small increase in wells after 2008 with a much larger increase after 2010.

2.2 HYDRAULIC FRACTURING AND HEALTH

The process of extracting oil and natural gas through both conventional and unconventional methods gives rise to health concerns, primarily through air and water pollution (Shonkoff, Hays, and Finkel 2014; Speight 2019). The use of large volumes of fracturing fluid, which contains potentially hazardous chemicals and can leach toxic substances from shale formations, distinguishes unconventional from conventional drilling. This water-intensive process contributes to concerns about drinking water contamination, and studies document elevated concentrations of potentially hazardous chemicals in drinking water near unconventional drilling sites (Alawattegama et al. 2015; Hill 2018; Rozell and Reaven 2012; U.S. EPA 2016; Wollin et al. 2020). In a review of the relationship between drinking water contamination, hydraulic fracturing, and health, Shonkoff, Hays, and Finkel (2014) cite studies that find elevated concentrations of methane (Jackson et al. 2013; Osborn et al. 2011), heavy metals and total dissolved solids (Fontenot et al. 2013), and volatile organic compounds (VOCs) including benzene, toluene, ethylbenzene, and xylenes (BTEX) (DiGiulio et al. 2011; Gross et al. 2013; Hildenbrand et al. 2015) in drinking water near unconventional drilling sites.

Exposure to these chemicals can have significant health consequences. For example, Webb et al. (2018)

Figure 1: Total wells drilled on Eagle Ford over time



Notes: Figure 1 shows the number of wells drilled each year-quarter in Eagle Ford during the sample period. ‘Total Wells’ includes all wells drilled during the period. ‘Horizontal Wells’ represents all horizontally drilled wells, which are the wells most likely to use hydraulic fracturing. Horizontal wells are a subset of the ‘Total Wells.’ *Source: Author’s Calculations using Texas Railroad Commission (2007–15) data.*

draws attention to the serious neurological health effects of chemicals used in the hydraulic fracturing process, such as heavy metals, particulate matter (PM), BTEX, and endocrine-disrupting chemicals (EDCs), all of which are associated with poor neurodevelopment, cognitive, and mental health outcomes through drinking water exposure. BTEX chemicals are also associated with damage to the nervous and endocrine systems, birth defects and low birth weight, respiratory conditions, abnormal heart rhythms, and blood diseases, including aplastic anemia and acute myelogenous leukemia (Epstein 2017). Shrestha et al. (2017) examine the health effects of chemicals found in groundwater near hydraulic fracturing sites, and find chemicals related to gastrointestinal distress, kidney disease or kidney damage, skin sensitivity or sores, cancer, dehydration, and liver disease, in addition to the conditions mentioned above. Chemicals used in drinking water treatment can also interact with chemicals used in hydraulic fracturing, transforming them into more toxic compounds. For instance, Wollin et al. (2020) explain that chemicals used to disinfect drinking water can cause oxidation of bromide used in hydraulic fracturing, creating the chemical bromate, which is mutagenic and carcinogenic.

Although the health effects of some chemicals used in the hydraulic fracturing process are known, there remains considerable uncertainty about both the substances used and their potential health consequences. Colborn et al. (2011) compile a non-exhaustive list of products used in the fracturing process and identify potential health effects for 353 chemicals for which Chemical Abstracts Service (CAS) numbers were available.

The chemicals with known health effects can impact the skin, eyes, kidneys, and brain as well as the nervous, immune, endocrine, respiratory, and cardiovascular systems. Twenty-five percent of the chemicals identified could cause cancer or mutations, and thirty-seven percent are candidate endocrine-disrupting chemicals (EDCs), which can affect reproductive systems and fetal and early childhood development. In a 2016 study on the impact of hydraulic fracturing on drinking water, the EPA finds that hydraulic fracturing can impact drinking water and considers the health effects of over 1,600 chemicals used in the hydraulic fracturing process. The authors note that the health effects of many of the chemicals used in the process are unknown and many are not regulated or measured in drinking water (U.S. EPA 2016).

In a cautionary public health paper, Finkel and Law (2011) explain that the timing of health effects after exposure to the chemicals used in hydraulic fracturing is largely understudied. For example, a review of clinical, observational, and epidemiological studies on EDCs suggests that numerous factors, such as age at exposure, exposure dosage, and mixtures of contaminants can all impact the timing between exposure and disease, and some endocrine-disrupting effects may appear decades after exposure or even transgenerationally. Prior studies also document adverse health outcomes in areas exposed to hydraulic fracturing, including higher rates of congenital heart defects among infants whose mothers lived near unconventional wells (McKenzie et al. 2014) and higher rates of hematological cancer among children living closer to oil and gas development (McKenzie et al. 2017).

Water contamination can affect both groundwater and surface water, but the spatial relationship between drilling and groundwater sources is more localized than the relationship between drilling and surface-water exposure. Rozell and Reaven (2012) identify several pathways through which unconventional drilling may affect water quality, including spills during fluid transportation, well-casing failures, subsurface migration, site discharge, and wastewater disposal. Surface-water exposure can depend on hydrologic features such as flow direction, elevation, and upstream activity (Bonetti, Leuz, and Michelon 2021; Hill and Ma 2022), making the relevant exposure area difficult to define geographically. By contrast, potential contamination of a groundwater source is more closely tied to drilling activity near the source location. Prior work suggests that contamination may arise from surface spills during the pre-production period (Shanafield, Cook, and Simmons 2019) or later through structural impairment, fluid migration, or discharge during flowback (Rozell and Reaven 2012; Shonkoff, Hays, and Finkel 2014). I therefore use proximity between drilling and public groundwater sources to examine whether the health effects of unconventional drilling vary with the location of population-serving water infrastructure. This measure captures the potential for a drinking-water pathway, but does not uniquely identify drinking-water exposure.

Air pollution provides an additional pathway through which unconventional drilling may affect health. Emissions arise from both drilling and production activities as well as the transportation of water, equipment,

and oil and natural gas to and from drilling sites. Prior studies document elevated concentrations of air pollutants near conventional and unconventional oil and gas development (McKenzie et al. 2012; Pétron et al. 2012; Roy, Adams, and Robinson 2014). Hydraulic fracturing also requires substantial truck traffic, which can increase diesel particulate matter along transportation routes (Shonkoff, Hays, and Finkel 2014; Speight 2019). Consistent with this pathway, Blundell and Kokoza (2022) use quasi-random variation in upwind natural gas flaring to identify a causal increase in respiratory-related hospital visits. Because these emissions can disperse beyond the immediate drilling site, the geographic footprint of air-pollution exposure may differ from the more localized relationship between drilling and groundwater-source locations. Drilling and production may also generate other localized exposures, such as noise, associated with industrial activity. The empirical analysis therefore does not separately identify a drinking-water pathway from other potential environmental pathways.

In addition to environmental exposures, shale development may affect healthcare utilization through economic and demographic channels. Shale booms can increase employment and wages and may alter insurance coverage, migration, and the composition of the local population. Inpatient care generally reflects more severe health needs than outpatient care and may therefore be less sensitive to short-run changes in income or employment (Hamad, Modrek, and Cullen 2016). Nonetheless, changes in population composition or insurance coverage could still affect measured inpatient utilization. I address these concerns empirically by examining utilization among women and among Medicare and Medicaid beneficiaries.

3 Data and Descriptive Statistics

3.1 HEALTHCARE DATA

Claim-level inpatient data from the Texas Health Care Information Collective (THCIC) provide the primary outcome data for this paper. All hospitals within the state of Texas are required to report claims, which are the invoices submitted to health insurance companies for payment, on discharged patients. THCIC provides de-identified claim reports available for public use (Texas Health Care Information Collective 2007–15).³ This rich data source includes information on each patient claim submitted by the hospital, the quarter and year the claim was submitted, details on the diagnosis and procedure codes associated with the claim, and patient demographic information.⁴ Specifically, the data include hospital information, such as the hospital name,

3. Although data were available through 2016 at the time of this study, healthcare providers switched from ICD-9 to ICD-10 codes in October of 2015. Therefore, I use all claims made before this transition.

4. The public use files exclude claims made in hospitals located in a county with a population less than 35,000, or those in a larger county with less than 100 hospital beds. In addition, hospitals that do not seek insurance or government reimbursement are also exempt.

indicators for the facility type,⁵ and categorical variables for the specialty unit⁶ where the care occurred. I also observe patient demographic information for each claim including the patient’s age range, gender, race, ethnicity, county, zip code, and payment information, such as insurance type. I use data between 2007 and 2015, starting one year prior to the entry of hydraulic fracturing in the Eagle Ford Shale formation. I also only include claims from individuals whose zip code can be assigned to a ZCTA in the treatment or comparison groups. If workers temporarily move to Eagle Ford for labor opportunities, but do not establish residency in the area, they are excluded from the analysis.

I categorize the ICD-9⁷ codes on each claim using the Healthcare Cost and Utilization Project’s Clinical Classifications Software (HCUP CCS), which collapses over 14,000 ICD-9 diagnosis codes and 3,900 procedure codes into clinically meaningful categories. Specifically, HCUP CCS categorizes diagnosis codes into a hierarchy of nested medical conditions. For example, a medical claim may include a set of diagnosis codes that suggest the clinically meaningful diagnosis was “Essential Hypertension,” which falls under the “Hypertension” category, which is nested under “Diseases of the Circulatory System.” For most conditions, I use the highest category in the hierarchy to test each medical category. I focus on health conditions that could be linked to hydraulic fracturing through pollution exposure according to prior literature described in Section 2.2. Specifically, I focus on mental health,⁸ endocrine conditions, cancer, conditions of the central nervous system, respiratory, gastrointestinal, urological, circulatory, and integumentary systems, pregnancy and birth outcomes, and blood diseases (Colborn et al. 2011; Epstein 2017; Shonkoff, Hays, and Finkel 2014; Webb et al. 2018; Wollin et al. 2020; Garshick et al. 2008; Persico and Marcotte 2022). In addition, I test changes in ‘General Illness,’ which includes symptoms such as fever, nausea and vomiting, and general malaise that is not attributed to a specific condition. This list is not exhaustive and serves as an analytical starting point for this study. For more information on the conditions included in this paper, see Appendix Section A.1.1.⁹

3.2 UNCONVENTIONAL DRILLING TECHNOLOGY

The data include several different types of wells and do not specifically identify hydraulic fracturing sites. For my analysis, I identify horizontally drilled, non-injection wells using the RRC horizontal-well classification,

5. For example, this might include an indicator for a psychiatric facility or a rehabilitation facility.

6. Specialty units are the areas of the hospital dedicated to patients with specific needs. For example, the obstetric unit is the area of the hospital specifically for labor and delivery.

7. ICD-9 and ICD-10 are the Ninth and Tenth Revisions, respectively, of the World Health Organization’s International Classification of Diseases, the official coding system used for diagnoses and procedures in the United States.

8. Mental health conditions include anxiety disorders, attention deficit disorders, depressive disorders, suicide and intentional self-harm, eating disorders, and substance abuse disorders.

9. I also tested the change in musculoskeletal conditions, such as injuries, as a potential placebo group. While I find no significant change in these conditions, they could plausibly vary with changes in local activity patterns, even when unrelated to water and air pollution. For example, Xu and Xu (2020) find an increase in fatal car accidents with the entry of a new hydraulic fracturing well. Therefore, I do not include a placebo category in this analysis.

and I refer to wells meeting these criteria as *qualifying wells*. I verify this classification using the surface and bottom well locations provided in the GIS data. Treatment is defined as the entry of the first qualifying well in a geographic area.

I map the Eagle Ford Shale boundary to identify wells located within two miles of the formation.¹⁰ Ultimately, I identify 17,965 qualifying wells drilled in the Eagle Ford region during the study period.

Treatment timing is determined by the spud date of the first qualifying well drilled in a geographic area beginning in the fourth quarter of 2008 (Blackmon 2018). This definition requires two key assumptions. First, because the data do not directly identify hydraulic fracturing, I use horizontal drilling in the Eagle Ford as a proxy for unconventional shale development. Speight (2019) states that oil and natural gas in the Eagle Ford Shale formation are inaccessible using conventional drilling techniques. Figure 1 shows that the increase in drilling activity in the region was driven almost entirely by horizontally drilled wells. Second, I assume that hydraulic fracturing occurs reasonably soon after the well is drilled. Conversations with engineers in the field suggest that this assumption is generally reasonable, although fracturing may sometimes be delayed. Because hydraulic fracturing stimulates oil and gas production, producers may delay completion until economic conditions, such as oil prices, are more favorable.

I conduct my analysis at the ZCTA level and identify treated Zip Code Tabulation Areas (ZCTAs)¹¹ using the boundaries of the shale formation. The estimation sample includes ZCTAs in the 27 Eagle Ford counties and the rural and semi-rural neighboring counties.¹² Within this sample, ZCTAs are treated if they receive a qualifying well during the study period and otherwise contribute to the never-treated comparison group. Treatment timing corresponds to the quarter when the first qualifying well was drilled within the ZCTA. This results in staggered treatment timing, with first treatment occurring between the fourth quarter of 2008 and the first quarter of 2015.

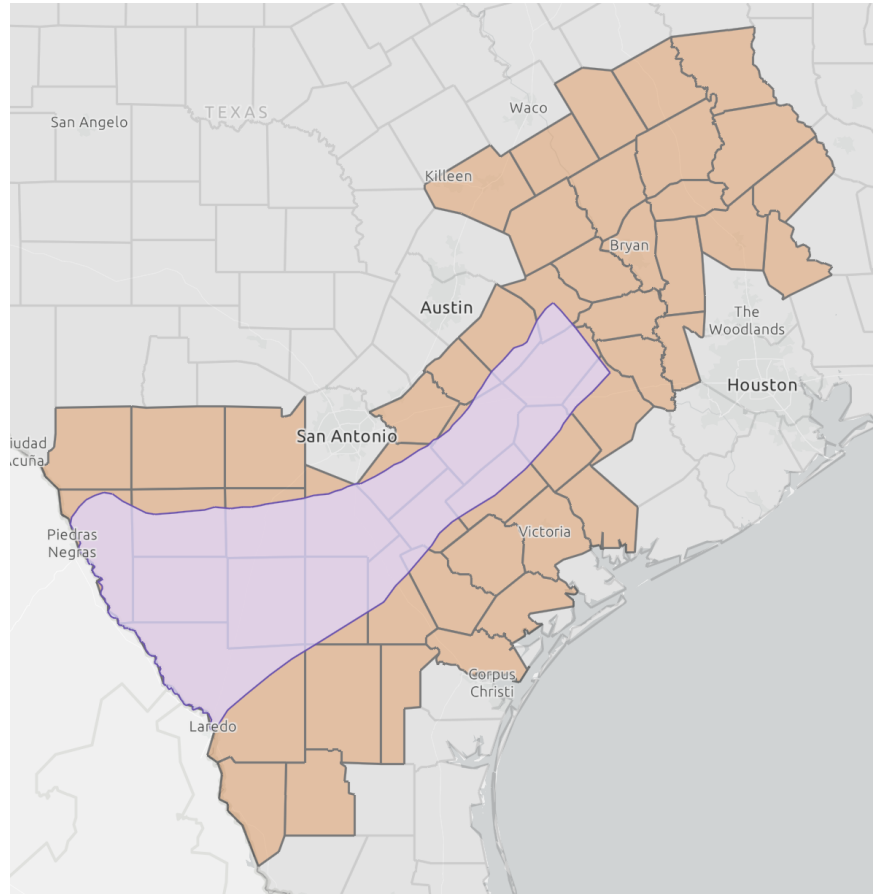
The drawback of conducting the analysis at the ZCTA level is that THCIC masks zip code identifiers for claims from patients living in zip codes with fewer than 30 patients total or fewer than five patients of a particular gender, making some utilization rates artificially low. Overall, approximately 11 percent of claims have suppressed zip code identifiers, although the suppression rate increases somewhat over the sample period and varies across conditions. Suppression is particularly low for pregnancy complications and perinatal conditions, at 2.7 percent and 1.0 percent of claims, respectively, while approximately 24 percent of mental-health claims have suppressed zip code identifiers. Estimates for mental-health conditions should

10. I choose a two-mile radius from the well location to the shale formation based on conversations with fracking engineers.

11. Zip codes are groups of street networks and buildings used by the United States Postal Service for mail delivery, while ZCTAs are designed by the United States Census Bureau for geographic analysis.

12. I exclude Bexar County (San Antonio), Travis County (Austin), Williamson County (Austin metropolitan area), and Montgomery County (Houston metropolitan area). These large metropolitan areas may have healthcare utilization and other economic or demographic dynamics that differ from the more rural Eagle Ford region.

Figure 2: Eagle Ford Shale formation



Notes: This figure shows the Eagle Ford Shale formation in southern Texas and highlights the Texas counties containing ZCTAs included in both the treatment and comparison groups. The purple shape overlaying the counties is the Eagle Ford Shale formation; treated ZCTAs include those containing at least one qualifying well within two miles of the shale formation boundary. *Source: Author's Calculations. Generated using ArcGIS Online.*

therefore be interpreted with greater caution.

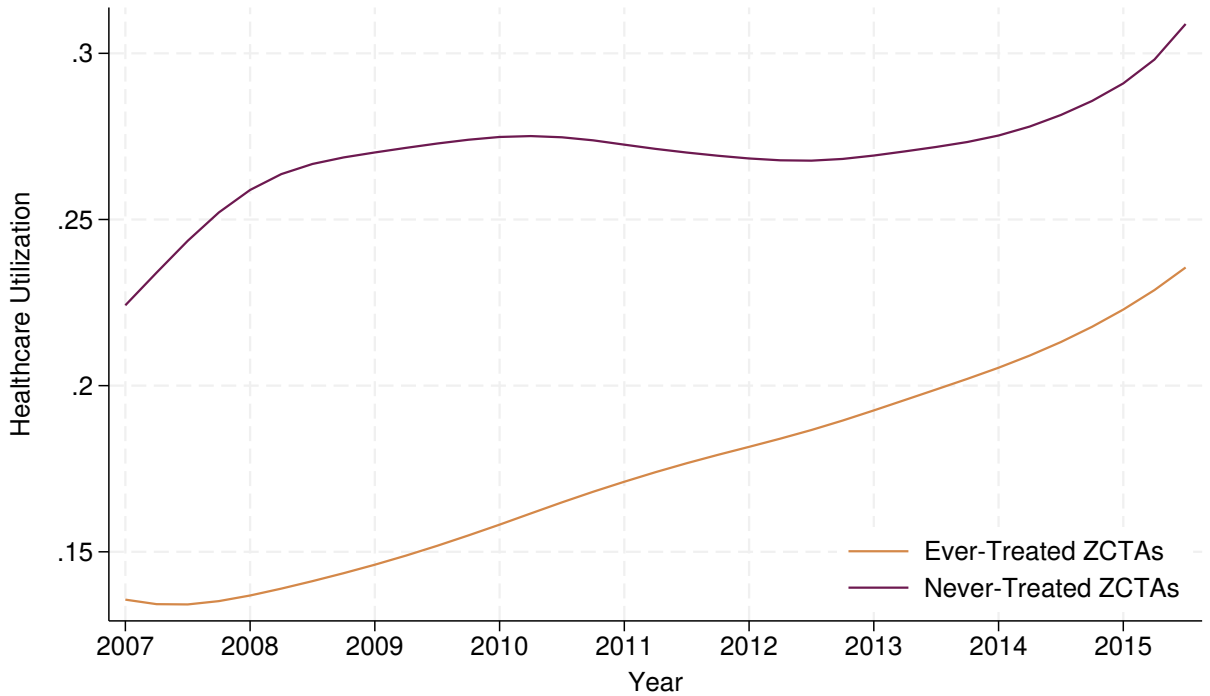
3.3 DESCRIPTIVE STATISTICS

My outcome variable of interest is the ZCTA-level healthcare utilization rate for each condition considered in this paper. I calculate the healthcare utilization rate as follows:

$$Y_{ct} \equiv \frac{\sum_{claim \in HC} claim_{ct}}{Pop_c} \quad (1)$$

Utilization Y_{ct} in ZCTA c in year-quarter t is the total number of claims, $claim_{ct}$, in healthcare category HC divided by the ZCTA population. Population is measured using the U.S. Census Bureau Decennial Census total at the ZCTA level.

Figure 3: Healthcare utilization in the treated and control ZCTAs



Notes: Figure 3 plots the average quarterly inpatient healthcare utilization for ever-treated and control ZCTAs. All ZCTAs with at least one qualifying well are the treated ZCTAs (in orange), while ZCTAs without qualifying wells (in purple) are the controls. The utilization rate for each healthcare category is calculated by taking the sum of all claims for the health conditions considered in this study made by residents of ZCTAs each quarter divided by the population of that ZCTA. *Source: Author's Calculations using data from Texas Health Care Information Collective (2007–15).*

Table 1 presents the average inpatient healthcare utilization rates in the baseline year of 2007 and across the overall panel for both the treated ZCTAs and the untreated ZCTAs. “All Conditions” represents the sum

of all claims made for conditions potentially related to hydraulic fracturing through pollution pathways, as described in Section 2.2. This includes circulatory conditions, general illness, mental health, endocrine conditions, cancer, central nervous system conditions, respiratory conditions, gastrointestinal issues, urological conditions, pregnancy complications, dermatological conditions, perinatal conditions, congenital abnormalities, and blood diseases.

Descriptively, never-treated ZCTAs tend to exhibit higher baseline inpatient healthcare utilization than ever-treated ZCTAs. However, in both the ever- and never-treated ZCTAs, the overall panel utilization rate is higher than in the baseline year. This suggests that inpatient utilization increased broadly for both groups throughout the study period, highlighting the importance of the staggered difference-in-differences approach to isolate the specific treatment effect from the general trend.

Figure 3 plots the average quarterly utilization rate for all conditions potentially related to hydraulic fracturing in the ever-treated ZCTAs and the never-treated ZCTAs, highlighting the general upward trend in the region. The never-treated ZCTAs exhibit greater fluctuation, including an initial increase between 2007 and 2010, then a slight decline between 2010 and 2012 before rising again. In contrast, utilization in the ever-treated ZCTAs increases smoothly and steadily throughout the entire period.

4 Methods and Results

4.1 HEALTHCARE UTILIZATION

I begin by estimating the effect of unconventional drilling entry within a ZCTA on healthcare utilization using a staggered difference-in-differences framework. I identify changes in the volume of inpatient claims from patients living in ever-treated ZCTAs ($N=49$) relative to the volume of claims from patients living in never-treated ZCTAs ($N=192$). ZCTAs are treated the year-quarter when the first qualifying well is drilled, and remain treated throughout the sample period. I use the estimator described in Callaway and Sant’Anna (2021), which exploits variation in treatment timing to calculate the group-time average treatment effects on the treated.

The comparison group includes ZCTAs within the counties surrounding the Eagle Ford Shale formation as well as never-treated and not-yet-treated ZCTAs on the shale formation. For each treatment cohort and year-quarter, the estimator captures the change in inpatient healthcare utilization among treated ZCTAs relative to the change among ZCTAs that have not yet experienced unconventional drilling nearby. I then aggregate these group-time effects across treatment cohorts and post-treatment periods to estimate the average effect of unconventional drilling entry on healthcare utilization among treated ZCTAs. This approach assumes

Table 1: Average ZCTA-Level Healthcare Utilization

	Baseline (2007)		Overall Panel	
	Ever-Treated	Never-Treated	Ever-Treated	Never-Treated
All Conditions	0.132 (0.096)	0.232 (0.708)	0.174 (0.108)	0.271 (0.862)
Cancer	0.006 (0.005)	0.010 (0.024)	0.008 (0.006)	0.011 (0.038)
Mental Health	0.011 (0.008)	0.021 (0.053)	0.014 (0.009)	0.023 (0.057)
Central Nervous System	0.009 (0.007)	0.017 (0.061)	0.013 (0.009)	0.019 (0.058)
Circulatory	0.039 (0.029)	0.064 (0.161)	0.049 (0.032)	0.076 (0.232)
Respiratory	0.013 (0.011)	0.024 (0.079)	0.018 (0.012)	0.029 (0.091)
Gastrointestinal	0.009 (0.007)	0.017 (0.055)	0.012 (0.008)	0.019 (0.070)
Urological	0.010 (0.007)	0.017 (0.070)	0.013 (0.008)	0.019 (0.077)
Pregnancy Complications	0.003 (0.003)	0.008 (0.047)	0.004 (0.003)	0.006 (0.023)
Dermatological	0.002 (0.002)	0.004 (0.013)	0.003 (0.002)	0.004 (0.021)
Perinatal	0.002 (0.002)	0.004 (0.016)	0.003 (0.003)	0.004 (0.018)
General Illness	0.004 (0.004)	0.007 (0.018)	0.006 (0.005)	0.008 (0.026)
Endocrine	0.026 (0.019)	0.043 (0.128)	0.036 (0.023)	0.057 (0.198)
Congenital Abnormalities	0.001 (0.001)	0.001 (0.002)	0.001 (0.001)	0.001 (0.006)
Blood Disease	0.006 (0.004)	0.010 (0.047)	0.008 (0.005)	0.012 (0.049)

Notes: Standard deviations are reported in parentheses. Utilization is calculated as the number of inpatient claims divided by the ZCTA population. Ever-treated ZCTAs receive at least one qualifying well during the study period. The baseline period reports utilization in 2007, before the onset of qualifying Eagle Ford drilling.

that, in the absence of unconventional drilling in the ZCTA, healthcare utilization in treated ZCTAs would have followed the same trends in utilization as the not-yet-treated and never-treated comparison groups.¹³

Table 2 displays the average treatment effect on the treated ZCTAs. The estimates reveal an increase in the overall volume of healthcare claims after the entry of qualifying drilling, although this effect is not statistically different from zero. I do find significant increases in pregnancy complications and perinatal conditions in ZCTAs with qualifying wells compared with those without. Prior studies, including Hill and Ma (2022) and Currie, Greenstone, and Meckel (2017), establish a link between unconventional drilling and adverse infant health outcomes. These studies use individual-level vital statistics and primarily examine outcomes observed at birth, such as birth weight and gestational age at birth. My results complement this literature by documenting an increase in inpatient utilization for pregnancy complications, capturing maternal health effects during pregnancy in addition to adverse perinatal outcomes.

The estimated increase in inpatient claims for pregnancy complications is approximately 0.0017 claims per resident per quarter, relative to a baseline rate of 0.0034, while the estimated increase in perinatal claims is approximately 0.0016 relative to a baseline rate of 0.0022. For an average ZCTA with a population of 6,792, the estimated increase in pregnancy-complication utilization corresponds to approximately 12 additional inpatient claims per quarter following the entry of qualifying drilling. The estimated increase in perinatal utilization similarly corresponds to approximately 11 additional claims per quarter.

One concern is that unconventional drilling may also affect fertility or population composition, which could mechanically increase the number of pregnancy-related inpatient claims even if the underlying risk of pregnancy complications is unchanged (Kearney and Wilson 2018). To assess this possibility, I estimate the same staggered difference-in-differences specification using the ZCTA-level birth rate as the outcome. The estimated effect is positive but not statistically significant: qualifying drilling is associated with an increase of approximately 0.0009 births per resident per quarter ($p=0.213$). In the average ZCTA, this corresponds to roughly 6 additional births per quarter. This increase is smaller than the estimated increase of approximately 12 pregnancy-complication-related inpatient claims per quarter, suggesting that the pregnancy-complication result is unlikely to be explained solely by an increase in births. However, births and inpatient claims are not directly comparable on a one-to-one basis, as a birthing person may have multiple complication-related claims per birth.

Finally, I find an increase in inpatient claims for dermatological conditions of approximately 6 additional claims per average ZCTA per quarter, which is significant at the 10% level. These increases in skin-related hospitalizations mirror findings by Denham et al. (2019), who find that the increases are driven by severe skin infections and abscesses.

13. See Figures A1 and A2 for event plots.

Table 2: Change in inpatient healthcare utilization following the first qualifying Eagle Ford well

Health Condition	Δ Utilization Rate			
	Full Sample (1)	Medicare (2)	Medicaid (3)	Female Patients (4)
All Conditions	0.0365 (0.0293)	0.0322 (0.0241)	0.0292 (0.0283)	0.0129 (0.0111)
Circulatory	0.0068 (0.0070)	0.0068 (0.0061)	0.0056 (0.0065)	0.0009 (0.0021)
General Illness	0.0015 (0.0013)	0.0009 (0.0008)	0.0012 (0.0012)	0.0010 (0.0010)
Mental Health	0.0042 (0.0028)	0.0016 (0.0013)	0.0028 (0.0021)	0.0007 (0.0006)
Endocrine	0.0058 (0.0055)	0.0068 (0.0051)	0.0060 (0.0057)	0.0029 (0.0028)
Cancer	0.0028 (0.0019)	0.0023 (0.0015)	0.0015 (0.0014)	0.0011 (0.0011)
Central Nervous System	0.0027 (0.0019)	0.0025 (0.0019)	0.0024 (0.0019)	0.0010 (0.0006)
Respiratory	0.0030 (0.0029)	0.0027 (0.0021)	0.0021 (0.0023)	0.0007 (0.0012)
Gastrointestinal	0.0033 (0.0026)	0.0035 (0.0025)	0.0029 (0.0030)	0.0019 (0.0018)
Urological	0.0040 (0.0029)	0.0035 (0.0023)	0.0029 (0.0030)	0.0010 (0.0008)
Pregnancy Complications	0.0017** (0.0008)	—	0.0009 (0.0006)	0.0017** (0.0008)
Dermatological	0.0009* (0.0005)	0.0010 (0.0006)	0.0009 (0.0007)	0.0002 (0.0002)
Perinatal	0.0016** (0.0008)	—	0.0011* (0.0006)	0.0003* (0.0002)
Blood Disease	0.0022 (0.0017)	0.0024 (0.0017)	0.0022 (0.0021)	0.0009 (0.0009)
Congenital Abnormalities	-0.0000 (0.0002)	—	-0.0000 (0.0001)	-0.0000 (0.0001)
Observations	8,435	8,435	8,435	8,435

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Notes: Table 2 reports aggregated Average Treatment Effects on the Treated from staggered difference-in-differences specifications estimated at the ZCTA-quarter level. Treatment begins in the quarter in which a ZCTA receives its first qualifying horizontal well located within two miles of the Eagle Ford Shale formation. The comparison group consists of not-yet-treated and never-treated ZCTAs.

Outcomes are measured as quarterly inpatient claims per total ZCTA resident. Column (1) uses claims for the full sample. Columns (2) and (3) count claims for which Medicare or Medicaid, respectively, is listed as either the primary or secondary payer. Because payer categories may overlap, the Medicare and Medicaid columns are not mutually exclusive. Column (4) includes inpatient claims for female patients. Pregnancy, perinatal, and congenital outcomes are omitted from the Medicare specification because they are not substantively relevant to the Medicare population. Estimates are weighted by ZCTA population. Standard errors are reported in parentheses.

For the remaining conditions, the estimates suggest general increases in the volume of inpatient claims, but are not statistically conclusive. Across all conditions, qualifying drilling is associated with an increase of approximately 0.036 inpatient claims per resident per quarter, corresponding to nearly 248 additional inpatient claims per quarter. However, the estimate is not distinguishable from zero.

I also estimate positive, though noisy, changes in utilization for conditions found to be associated with hydraulic fracturing in other literature, including respiratory and circulatory conditions (Peng, Meyerhoefer, and Chou 2018; Xu 2023). I estimate an increase of approximately 46 additional inpatient claims per average ZCTA each quarter. Peng, Meyerhoefer, and Chou (2018) also report evidence of increased acute myocardial infarction hospitalizations associated with unconventional natural gas development, although these estimates are less robust than their primary findings for pneumonia. My estimates for circulatory conditions, which include acute myocardial infarctions, align with the direction of the results in Peng, Meyerhoefer, and Chou (2018), but do not rise to the level of statistical significance. My circulatory outcome includes a broad set of diagnoses, which may make it more difficult to detect effects concentrated in a narrower subset of conditions. However, if pollution-related impacts are concentrated in a subset of conditions, the estimates based on narrower outcomes may yield larger estimated effects relative to broader measures of circulatory health.

Similarly, I also find that inpatient claims for respiratory conditions increase by approximately 20 additional claims per average ZCTA per quarter, which is also indistinguishable from zero. Both Peng, Meyerhoefer, and Chou (2018) and Xu (2023) find significant increases in respiratory hospitalizations for pneumonia and asthma, respectively, following unconventional drilling entry. As with circulatory conditions, the respiratory conditions captured in this paper include a broad set of conditions, such as acute infections (e.g., pneumonia, influenza, bronchitis), chronic respiratory diseases (e.g., COPD and asthma), and severe respiratory outcomes (e.g., respiratory failure and pleural conditions), which may explain the difference in the results.

Differences in the populations and settings captured by these studies may also contribute to the contrasting results. Air-pollution-related respiratory symptoms may not result in inpatient admission for all patients. Children and older adults are particularly vulnerable to severe respiratory illness and may be more likely to require hospital care (Rasmussen, Ogburn, McCormack, et al. 2016; Strosnider et al. 2019; Willis et al. 2018; Bell, Dominici, et al. 2013). Thus, the asthma-related effects among children documented by Xu (2023) and the pneumonia effects among older adults documented by Peng, Meyerhoefer, and Chou (2018) may be more readily detectable than changes in broad respiratory inpatient utilization across the full population.

4.2 COMPOSITIONAL CHANGES

One concern about comparing Eagle Ford areas to the border areas is that the population composition may change in response to new opportunities related to oil and natural gas production. Therefore, the estimated health effects may be capturing changes in the composition of the local population, rather than the negative effects of pollution exposure to bystanders. Wilson (2022) estimates that individuals who move for fracking-related employment opportunities account for an increase of less than two percent of the total population in fracking counties in the southern United States. These estimates suggest that migration is unlikely to be the primary driver of the observed effects, though it may contribute modestly. Alternatively, the effects captured in this analysis may be related to income effects from the fracking boom, including increases in individuals with health insurance, although focusing on inpatient claims partially mitigates this concern.

Demographic groups that move to fracking counties tend to be unmarried men under the age of 65 (Willis et al. 2018). To address concerns about compositional changes, I re-estimate the effect of unconventional drilling on inpatient claims for women and for Medicaid or Medicare beneficiaries, who are less likely to be affected by fracking-related migration. Focusing on these subpopulations also helps address concerns about changes in insurance coverage, since Medicaid and Medicare are government-provided rather than employer-sponsored plans. If the results in Column (1) were driven primarily by an influx of young men or changes in employer-sponsored insurance coverage, I would expect the estimates to differ more substantially across these groups.

Columns (2)–(4) of Table 2 display the results for these subpopulations. Among Medicare beneficiaries, who are primarily adults age 65 and older, the estimated effect on overall healthcare utilization is positive and similar in magnitude to the full-sample estimate, although, like the full sample estimates, it is not statistically different from zero. The estimated increase of 0.032 claims per resident corresponds to approximately 219 additional Medicare-paid claims per ZCTA-quarter in the average ZCTA. Estimates across individual health conditions are also generally positive, though imprecisely estimated.

The estimated effect among Medicaid beneficiaries is also positive and comparable in magnitude to the full-sample estimate. The estimated increase of 0.029 claims per resident corresponds to approximately 198 additional Medicaid inpatient claims per ZCTA-quarter. I also find a positive increase in Medicaid claims for perinatal conditions, corresponding to approximately 8 additional Medicaid claims per ZCTA-quarter, which is statistically significant at the 10 percent level. Because Medicaid primarily serves low-income populations and is less directly tied to local employer-sponsored insurance, this analysis suggests that changes in private insurance coverage are not driving the results.

Finally, Column (4) of Table 2 reports the estimates for the female subpopulation. Unlike the Medicare

and Medicaid cohorts, the estimated increase in healthcare utilization is substantially smaller than the full-sample estimate. However, the female estimates remain positive across several outcomes. Together with the similar estimates among Medicare and Medicaid beneficiaries, these subgroup results suggest that changes in population composition alone are unlikely to explain the full-sample findings, although they cannot fully distinguish compositional changes from heterogeneous health effects.

As part of this broader set of robustness checks, I also restrict the sample to ZCTAs located within the 27 Eagle Ford counties, excluding neighboring counties from the analysis. In this more geographically narrow specification, the estimated effect on overall inpatient utilization is close to zero and not statistically distinguishable from zero. However, the estimates for pregnancy complications and perinatal conditions remain positive. The estimated increase in pregnancy-complication utilization is approximately 0.0009 claims per resident per quarter ($p=0.061$), while the estimated increase in perinatal utilization is approximately 0.0008 claims per resident per quarter ($p=0.026$). These results suggest that the maternal and perinatal findings are less sensitive to the choice of geographic comparison group than the broader utilization effect.

4.3 WATER CONTAMINATION

To evaluate whether the health effects in Section 4.1 are consistent with a drinking-water pathway, I examine how healthcare utilization changes when qualifying drilling occurs near a public groundwater source serving the local population. This comparison also reduces concerns that the estimates are driven solely by broader population changes associated with shale development. I focus on groundwater sources rather than surface-water sources because potential contamination from spills, leaks, discharge, or wastewater disposal is more spatially localized around the drilling site, whereas surface-water contamination may depend on more complex upstream and downstream dynamics.

Using data from the Texas Water Development Board (TWDB) and ZCTA boundaries, I identify the ZCTAs served by each PWSA as those that intersect the PWSA service area. I then measure whether a qualifying well is drilled within one, three, or five kilometers of a public groundwater source used by the PWSA serving that ZCTA. Treatment is defined at each distance as the first quarter in which qualifying drilling occurs within the specified radius. Using the estimator in Callaway and Sant’Anna (2021), I compare treated ZCTAs to ZCTAs that also experience qualifying unconventional drilling during the sample period, but whose public groundwater source locations are more than five kilometers from qualifying drilling. I restrict the analysis to ZCTAs that receive water from a public groundwater system, thereby excluding ZCTAs relying on private water sources.

These measures capture potential exposure through a drinking-water pathway, although I do not directly

observe groundwater contamination. Because the comparison group also experiences qualifying shale development, this specification compares the health effects of drilling located near population-serving water infrastructure with the health effects of drilling elsewhere in communities that experience shale development. Therefore, the estimation strategy isolates the additional effect associated with drilling near drinking-water infrastructure, beyond the broader effect of drilling in the community.

Table 3 presents estimates of the change in inpatient healthcare utilization associated with qualifying drilling near public groundwater sources. Event studies corresponding to the five-kilometer estimates are shown in Figures A3 and A4. In general, the results are more precisely estimated in the five-kilometer specification than in the one-kilometer specification. At the five-kilometer distance, drilling near a public groundwater source is associated with a statistically significant increase of 0.0338 inpatient claims per resident per quarter. The average population of a ZCTA with drilling at the five kilometer level is 4,977 residents, and therefore the estimates translate to an increase of approximately 168 inpatient claims per ZCTA-quarter.

I also find positive and statistically significant increases for individual conditions, including circulatory, mental health, cancer, central nervous system, respiratory, gastrointestinal, and general illness diagnoses. Many of these estimates are similar or slightly smaller than the corresponding estimates in the primary analysis, but are more tightly estimated. For example, the estimated effect for cancer is 0.0028 claims per resident per quarter in the general analysis and 0.0026 in the five-kilometer groundwater specification. While the estimated rates are similar, the implied increase in claims for cancer translates to nearly 13 claims per ZCTA-quarter in the groundwater analysis, smaller than the implied 19 claims per ZCTA-quarter in the primary analysis. Similarly, the estimates for gastrointestinal conditions are 0.0033 in the primary analysis and 0.0030 in the groundwater analysis which translates to approximately 22 and 15 additional claims per ZCTA-quarter, respectively. Taken together, these comparisons suggest that the significant groundwater estimates reflect greater precision within this more targeted sample, rather than substantially larger estimated effects than those observed in the primary analysis. The groundwater-source specification is therefore informative about whether the effects of unconventional drilling vary with proximity to public drinking-water infrastructure, but it does not identify the specific environmental pathway responsible for those effects. The estimates may reflect drinking-water exposure, other localized exposures associated with drilling, or a combination of pathways.

Perinatal outcomes and pregnancy complications exhibit a more localized pattern, with larger estimates at shorter distances from public groundwater sources. The result for perinatal outcomes is statistically significant at the three-kilometer level; I find that drilling within three kilometers of a groundwater site is associated with 0.0006 additional perinatal claims, or about three additional claims per ZCTA-quarter. These estimates are consistent with the broader literature documenting adverse infant health effects associated with

Table 3: Change in Healthcare Utilization Associated with Drilling Near Public Groundwater Sources

Health Condition	≤ 1 km (1)	≤ 3 km (2)	≤ 5 km (3)
All Conditions	0.0045 (0.0153)	0.0260 (0.0166)	0.0338** (0.0146)
Circulatory	0.0015 (0.0042)	0.0077 (0.0051)	0.0100** (0.0047)
General Illness	0.0005 (0.0006)	0.0016** (0.0006)	0.0015** (0.0007)
Mental Health	0.0011 (0.0012)	0.0029** (0.0012)	0.0034*** (0.0011)
Endocrine	0.0003 (0.0030)	0.0039 (0.0037)	0.0055* (0.0032)
Cancer	-0.0006 (0.0011)	0.0015* (0.0009)	0.0026*** (0.0008)
Central Nervous System	0.0002 (0.0018)	0.0020 (0.0014)	0.0023** (0.0011)
Respiratory	0.0001 (0.0018)	0.0029* (0.0017)	0.0041*** (0.0015)
Gastrointestinal	0.0003 (0.0014)	0.0026* (0.0014)	0.0030** (0.0014)
Urological	0.0000 (0.0016)	0.0009 (0.0014)	0.0014 (0.0012)
Pregnancy Complications	0.0012** (0.0006)	0.0008 (0.0005)	0.0004 (0.0005)
Dermatological	-0.0006 (0.0005)	0.0002 (0.0004)	0.0005 (0.0004)
Perinatal	0.0007 (0.0005)	0.0006** (0.0003)	0.0004 (0.0004)
Blood Disease	0.0003 (0.0009)	0.0007 (0.0009)	0.0011 (0.0008)
Congenital Abnormalities	0.0002 (0.0001)	0.0000 (0.0001)	-0.0000 (0.0001)
Treated ZCTAs	21	33	35
Control ZCTAs	13	13	13
Observations	1,190	1,610	1,680

Standard errors in parentheses.

*Significant at the 10 percent level; **Significant at the 5 percent level;

***Significant at the 1 percent level.

Notes: Estimates use the Callaway and Sant’Anna (2021) estimator and are weighted by ZCTA population. The sample is restricted to ZCTAs in the Eagle Ford region that are served by public water systems and experience qualifying horizontal Eagle Ford drilling. Treatment occurs when a public groundwater source serving the ZCTA is first located within the specified distance (≤ 1 km, ≤ 3 km, or ≤ 5 km) of a qualifying well. For each specification, the comparison group consists of drilling ZCTAs whose public groundwater sources remain more than 5 km from qualifying drilling.

unconventional drilling within three kilometers of the mother’s residence (Currie, Greenstone, and Meckel 2017; Hill and Ma 2022).

In contrast, pregnancy complications capture a distinct dimension of health: maternal morbidity during pregnancy. I find that qualifying drilling is associated with approximately 12 additional pregnancy-complication claims per ZCTA-quarter in the primary specification. In the groundwater analysis, drilling within one kilometer of a public groundwater source is associated with approximately 0.0012 additional pregnancy-complication claims per resident per quarter, or about 7 additional claims in the average treated ZCTA.

The persistence of the pregnancy-complication estimate across the primary analysis and the narrowest groundwater specification suggests that the health consequences of unconventional drilling likely extend beyond infant outcomes to maternal health during pregnancy. Although the literature on maternal health outcomes related to unconventional drilling is less well-developed than the literature on infant health outcomes, prior work identifies a relationship between hydraulic fracturing entry and poor maternal health. Most notably, Willis et al. (2022) use a difference-in-differences design in Texas and find that residential proximity to active oil and gas extraction increases the risk of gestational hypertension and eclampsia. Other epidemiologic studies associate unconventional natural gas development with higher odds of mental illness and substance use among pregnant people (Aker, Whitworth, et al. 2022).

Finally, I estimate changes in the concentration of water contaminants before and after the entry of a qualifying well near public water infrastructure using groundwater-contamination data from the [Texas Commission on Environmental Quality Drinking Water Watch](#) (Texas Commission on Environmental Quality 2024). I find a positive and statistically significant increase in total dissolved solids of 25.5 units, equivalent to approximately 5.6% of the estimation-sample mean. I also separately examine contaminants associated with unconventional drilling (Hill and Ma 2022) and find evidence of broader changes in groundwater chemistry. Calcium and magnesium, both associated with produced water, increase by roughly 31% and 30%, respectively, relative to their estimation-sample means. Nitrate and fluoride, which are also associated with produced water, increase significantly in at least one specification. Sulfate and chloride, which are associated with both fracturing fluid and produced water, likewise show statistically significant increases in at least one specification. In contrast, iron concentrations fall significantly following the entry of nearby drilling.

Importantly, monitoring schedules for drinking-water contaminants vary considerably across contaminants. Some are tested frequently, while others are tested annually or every three years. If contamination occurs through surface spills during the pre-production period, irregular testing around the timing of nearby drilling may cause short-lived changes in water quality to be missed. I therefore interpret these results as suggestive evidence that drilling near public groundwater sources alters measured groundwater chemistry

rather than as a definitive measure of drinking-water contamination. This interpretation is consistent with prior research finding that unconventional drilling within one kilometer of a water source increases concentrations of contaminants associated with hydraulic fracturing (Hill and Ma 2022). Additional details are provided in Appendix Section A.2.

5 Discussion and Conclusion

The past two decades saw dramatic increases in hydraulic fracturing and horizontal drilling technologies used in the production of oil and natural gas, leading to changes in environmental quality in shale regions. This paper estimates how the expansion of unconventional drilling affected healthcare utilization in the Eagle Ford Shale region of southern Texas.

I first examine changes in overall healthcare utilization following the entry of unconventional drilling. The point estimate implies approximately 248 additional inpatient claims per ZCTA-quarter, although the aggregate estimate is not statistically distinguishable from zero.

I do find significant increases in perinatal claims, pregnancy complications, and dermatological claims. Specifically, perinatal claims increase by 11 additional claims per ZCTA-quarter, pregnancy complications increase by 12 additional claims per ZCTA-quarter, and dermatological claims increase by 6 additional claims per ZCTA-quarter. The pregnancy-related estimates are substantial relative to baseline utilization: pregnancy-complication claims increase by approximately 50 percent and perinatal claims by approximately 73 percent. Although these utilization measures are not directly comparable to prior literature, which relies on individual birth outcomes, the magnitudes reinforce evidence from Currie, Greenstone, and Meckel (2017) and Willis et al. (2022) that unconventional drilling can generate meaningful health consequences during pregnancy and infancy.

To account for population-level changes, I repeat the specification for subpopulations that are less likely to be impacted by male-dominated compositional changes or changes in employer-sponsored insurance coverage. The estimates remain positive across Medicare, Medicaid, and female patients, although the aggregate estimate is smaller for women. The consistency in direction across these groups makes compositional changes alone a less compelling explanation for the observed increases.

I also provide causal evidence that the health effects of unconventional drilling depend on where drilling occurs relative to public groundwater infrastructure. Unlike the primary analysis, which compares ZCTAs with and without qualifying drilling, this specification restricts the sample to communities that all experience unconventional drilling and distinguishes exposure by proximity to public groundwater sources. ZCTAs that experience qualifying drilling within five kilometers of a public groundwater source have significant increases

in inpatient healthcare utilization relative to ZCTAs where drilling occurs farther from those sources. The aggregate estimate corresponds to approximately 168 additional claims per ZCTA-quarter, with increases in circulatory, mental health, cancer, central nervous system, gastrointestinal, respiratory, and general illness claims. These estimates are generally similar in magnitude to those in the primary analysis but are more precisely estimated. This within-drilling comparison provides evidence that the location of drilling matters for healthcare utilization. Because the design does not isolate the specific exposure pathway, these effects may reflect drinking-water contamination, other localized environmental exposures associated with drilling, or both.

The public water infrastructure analysis also reveals spatially concentrated increases in perinatal claims and pregnancy complications that differ from the pattern observed for other outcomes. Qualifying drilling near public water infrastructure is associated with approximately 3 additional perinatal claims at the three-kilometer threshold and 7 additional pregnancy-complication claims at the one-kilometer threshold. This result is consistent with prior literature documenting adverse infant health effects from unconventional drilling within three kilometers of maternal residence, while the pregnancy-complication result extends that literature by identifying maternal morbidity during pregnancy.

Although this paper finds a relationship between unconventional drilling and healthcare utilization, there are several limitations. The public-water analysis identifies the causal effect of drilling near public drinking-water infrastructure relative to drilling farther from that infrastructure, but it does not separately identify the environmental pathway responsible for that effect. The estimated effects could reflect exposure to water contamination, air pollution, noise pollution, or other exposures. Research identifying the causal impact of these alternative pathways on adult health conditions would provide a more complete picture of the effects of hydraulic fracturing on human health.

A second limitation is that the health-condition categories are necessarily broad. I select conditions based on prior medical and public-health evidence linking them to pollutants associated with unconventional drilling, but these categories encompass heterogeneous diagnoses and may mask effects concentrated in particular conditions. Future work using more clinically specific outcomes could better identify which dimensions of health are most affected.

These findings have implications for the regulation and monitoring of unconventional drilling. Because I find additional health effects when drilling occurs near public groundwater infrastructure, regulators may want to consider proximity to population-serving water sources when evaluating drilling locations and setback requirements. The results also highlight the value of systematic groundwater monitoring before and after drilling begins, particularly near public water sources, both to identify potential contamination and to better distinguish drinking-water exposure from other localized environmental pathways. Public-health

surveillance in shale regions could also benefit from greater attention to adult healthcare utilization and maternal morbidity, particularly given the persistence of pregnancy-complication effects across specifications.

More broadly, these results identify healthcare utilization as an additional external cost of unconventional drilling. The consumer benefits generated by shale development are substantial (Hausman and Kellogg [2015](#)), but a complete welfare assessment should also incorporate health costs borne by nearby residents. My findings suggest that these costs depend not only on whether drilling occurs, but also on where drilling is located relative to population-serving infrastructure.

6 References

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Appendix A

A.1 Data Description

This paper combines several sources of data, and in this section, I provide links to publicly available data sources and describe how I merge the sources to create the dataset used in the analysis.

A.1.1 Health Outcome Data

Publicly available inpatient healthcare discharge data from Texas Health Care Information Collective ([2007–15](#)) provide quarterly details about inpatient claims, along with the patient’s county and zip code of residence, which I use to create the ZCTA panels. Additionally, the data include the health insurer and patient gender, which I use for the heterogeneity analysis. I use data covering 2007 to the third quarter of 2015. The data are available for download on the [THCIC webpage](#).

I classify the healthcare conditions using the [HCUP Multi-Level CCS software](#), which takes the diagnosis and procedural codes and categorizes them into clinically meaningful categories. The table below breaks down the CCS codes that I include for each condition.

Table A1: HCUP CCS Codes Used in this Analysis

Condition Name	L1 CCS Codes	L2 CCS Codes
Circulatory	7	All codes
General Illness	17	17.1
Mental Health	5	All codes
Endocrine	3	All codes
Cancer	2	All codes
Central Nervous System	6	All codes
Respiratory	8	All codes
Gastrointestinal	9	All codes
Urological	10	All codes
Pregnancy Complications	11	11.2, 11.3, 11.4, 11.5, 11.6
Dermatological	12	All codes
Perinatal	15	15.2, 15.3, 15.4, 15.5, 15.6, 15.7
Blood Disease	4	All codes
Congenital Abnormalities	14	All codes

A.1.2 OIL AND NATURAL GAS PERMIT DATA

To identify the location of oil and gas wells, I use public data from the [RRC Drilling Permit Master File](#) and the [RRC GIS data files](#). I exclude permits that are missing location or spud date data, and I keep all permits with a spud date between 2007 and 2015.

I use ArcGIS to match well locations to the county and ZCTA boundaries from the U.S. Census Bureau using the Spatial Join tool. I identify which wells were drilled within two miles of the boundary of the Eagle Ford Shale formation using shapefiles from the U.S. Energy Information Administration, again using a Spatial Join.

Finally, I use the ArcGIS Near Table tool to identify qualifying wells drilled within one, three, and five kilometers of a public groundwater source. I then use the spud dates from the Permit Master File to determine the treatment timing.

A.1.3 WATER SOURCE DATA

Groundwater data, including the latitude and longitude of the water source, volume of surface and groundwater used, and number of individuals served were privately requested and compiled for this project by the [Texas Water Development Board](#). As described above, I use the latitude and longitude of the PWS location from these data to match nearby oil and natural gas wells using the Near Table tool in ArcGIS.

A.1.4 ADDITIONAL DATA

I also use data from the U.S. Census Bureau Population Totals and the Decennial Census totals ([U.S. Census Bureau 2024](#)) for the total population of the counties and ZCTAs used in the analysis, as well as descriptive statistics comparing demographics of the counties.

A.2 Groundwater Contaminant Analysis

Using data purchased from the [Texas Commission on Environmental Quality’s Texas Drinking Water Watch](#), I examine whether qualifying unconventional drilling near a public groundwater source is associated with changes in measured groundwater chemistry. These data include the date and PWS where the sample was collected; the sample point (distribution or entry point); the analysis method; contaminant names; and test results. This allows me to compare contaminant concentrations before and after qualifying drilling occurs near a public groundwater source, including concentrations below the maximum threshold set by the Safe Drinking Water Act.

An accurate measure of the relationship between qualifying drilling near public water infrastructure and drinking water quality would require testing the source water at the affected and unaffected sites regularly for the exact contaminants used in the hydraulic fracturing process, including testing before and after drilling. However, the monitoring schedule for different drinking water contaminants varies considerably depending on the contaminant in question. The variation in testing frequency makes it challenging to precisely identify changes in contaminant concentrations around the unconventional drilling entry date. Therefore, I estimate the average change in concentrations of contaminants for PWSs with unconventional drilling within one kilometer compared to PWSs without unconventional drilling nearby. I use the following two-way fixed effects specification:

$$Y_{pt} = \beta_0 + \beta_1 Treat_{pt} + \eta_p + \tau_t + \epsilon_{pt} \quad (2)$$

where Y_{pt} is the test result for each contaminant at PWS p measured at time t . $Treat_{pt}$ is an indicator equal to one in all periods after the first qualifying unconventional well is drilled within one kilometer of PWS p . The specification therefore compares changes in measured water chemistry after nearby drilling begins with contemporaneous changes at PWSs that remain farther from qualifying drilling. I restrict the comparison group to PWSs that never experience qualifying drilling within five kilometers during the sample period. PWSs that experience qualifying drilling between one and five kilometers are excluded from this analysis. These sources also may experience an increase in hydraulic fracturing chemicals during this time period, which would bias the results toward zero. To account for seasonality that may impact water contamination, I include year-quarter fixed effects, τ_t . I also include a fixed effect for each PWS, η_p . Standard errors are clustered at the PWS level. β_1 estimates the average change in a contaminant level within PWS-contaminant groups, measured in micrograms per liter ($\mu g/L$), after a qualifying well is drilled within one kilometer of a PWS.

Panel A of Table A2 shows changes in the level of total dissolved solids (TDS) and the aggregate measure of

fracking-related contaminants. TDS increases by approximately 25.5 units following qualifying drilling within one kilometer, equivalent to about 5.6% of the estimation-sample mean, and the estimate is statistically significant. I also construct an aggregate measure of contaminants potentially associated with produced water. The estimated change in this aggregate measure is positive but imprecisely estimated and is not statistically different from zero.

Many contaminants are not detectable below a reporting threshold, resulting in zero or non-positive values that cannot be included in a log specification. In Panel B of Table A2, I therefore estimate the same specification using the log of measured concentrations among observations with positive values. The log estimate for TDS is similar in direction and implies an increase of approximately 7.8%, and is statistically different from zero.

I test each contaminant related to hydraulic fracturing either through fracturing fluid or produced water and include the results in Table A3. Calcium, magnesium, nitrate, and fluoride—all associated with produced water—increase significantly in at least one specification. Sulfate and chloride, which are associated with both fracturing fluid and produced water, also increase significantly in at least one specification. In contrast, iron, another contaminant associated with both fracturing fluid and produced water, falls significantly following the entry of nearby drilling. Taken together, the contaminant-level estimates indicate changes in measured groundwater chemistry following nearby drilling, but the mixed signs and irregular monitoring schedule prevent a clean attribution to a specific contamination pathway.

Table A2: The effects of unconventional drilling near a public groundwater source on water chemistry

	Estimate	Observations	Mean
Panel A: Level Changes			
Total Dissolved Solids	25.53** (11.93)	2,471	455.90
All Fracking-Related Chemicals	191.94 (278.98)	172,328	3,537.27
Panel B: Log Changes			
Log(Total Dissolved Solids)	0.075*** (0.022)	2,471	
Log(All Fracking-Related Chemicals)	-0.211 (0.138)	78,507	

Standard errors in parentheses, clustered at the public water system level.

* Significant at the 10 percent level; ** significant at the 5 percent level; *** significant at the 1 percent level.

Notes: This table reports changes in measured water chemistry after the first qualifying unconventional well is drilled within 1 km of a public groundwater source. Treated public water systems are compared with systems that never experience qualifying drilling within 5 km during the sample period. All specifications include public water system and year-quarter fixed effects. Panel A reports changes in contaminant concentrations in levels. The Mean column reports the mean of the dependent variable in the corresponding level-regression estimation sample. Panel B reports changes in log concentration. The aggregate fracking-related-chemicals measure includes the predefined set of contaminants identified as potentially associated with hydraulic fracturing or produced water.

Table A3: Changes in groundwater contaminants following nearby unconventional drilling

Water Contaminant	Level Estimate	Mean	Obs. (Levels)	Log Estimate	Obs. (Logs)
Lead (1030)	-0.185 (0.241)	1.610	15,981	-0.063 (0.207)	9,444
Bromodichloromethane (2943)	0.194 (1.231)	8.353	11,257	-0.051 (0.170)	8,129
Copper (1022)	8.259 (17.673)	119.626	15,740	0.151 (0.213)	14,278
Xylenes, total (2955)	-0.586 (0.421)	0.396	2,602	-0.503 (0.367)	226
Nitrate (1040)	89.079** (36.059)	538.844	7,427	-0.056 (0.084)	5,533
Arsenic (1005)	-0.773 (2.171)	9.015	2,524	-0.053 (0.258)	1,450
Chloroform (2941)	0.736 (0.770)	7.965	11,257	-0.132 (0.130)	7,352
Bromoform (2942)	1.075 (1.201)	6.494	11,257	0.090 (0.083)	7,761
Fluoride (1025)	99.570** (42.744)	510.316	2,553	0.204 (0.126)	2,401
Barium (1010)	-11.272* (6.706)	169.533	1,716	-0.052 (0.104)	1,715
Nickel (1036)	-0.286 (0.181)	0.949	1,670	-0.010 (0.108)	862
Selenium (1045)	-0.391 (0.554)	1.431	1,671	-0.335 (0.487)	484
Radium-228 (4030)	-0.067 (0.203)	0.505	1,739	-0.125* (0.066)	405
Radium-226 (4020)	0.777 (1.936)	3.711	659	0.154 (0.287)	522
Manganese (1032)	-6.069* (3.567)	19.170	1,730	-0.239 (0.404)	1,520
Iron (1028)	-170.551*** (47.332)	172.907	1,746	-0.381 (0.368)	1,271
Sodium (1052)	-4,872.487 (5,134.191)	116,741.990	1,769	-0.073* (0.041)	1,769
Calcium (1016)	11,198.978*** (3,357.841)	35,978.940	1,668	0.306*** (0.108)	1,640
Chloride (1017)	3,802.916 (4,360.822)	82,403.892	2,474	0.153*** (0.037)	2,468
Sulfate (1055)	11,116.313** (5,638.871)	46,768.209	2,474	0.152*** (0.047)	2,257
Aluminum (1002)	16.205 (18.073)	19.841	1,670	1.357*** (0.454)	468
Zinc (1095)	-5.510 (25.236)	56.313	1,670	0.288* (0.162)	1,358
Magnesium (1031)	2,031.337** (1,002.114)	6,701.759	1,668	0.122 (0.075)	1,418
Potassium (1042)	—	5,136.398	450	—	443

Notes: Each row reports a separate fixed-effects regression estimating the change in the measured constituent after the first qualifying unconventional well is drilled within 1 km of a public groundwater source. Treated public water systems are compared with systems that never experience qualifying drilling within 5 km during the sample period. All specifications include public water system and year-quarter fixed effects. Standard errors, reported in parentheses, are clustered at the public water system level. The Mean column reports the mean of the dependent variable in the corresponding level-regression estimation sample. Log specifications use the log of measured concentration and therefore may contain fewer observations than the corresponding level specification. “—” indicates that the treatment coefficient is not identified. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

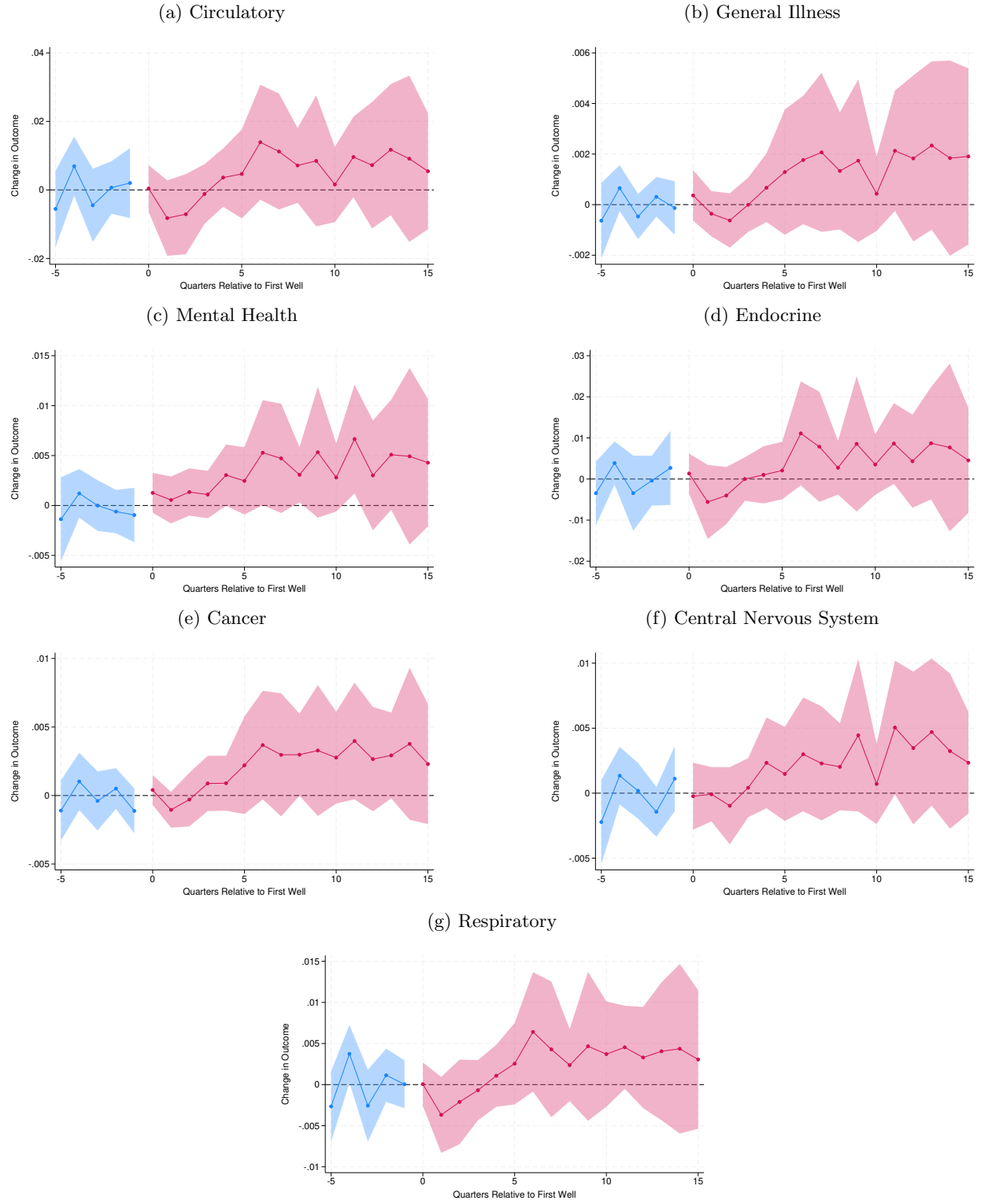
A.3 Event Study Figures

Figure A1: Fracking-related healthcare utilization event study



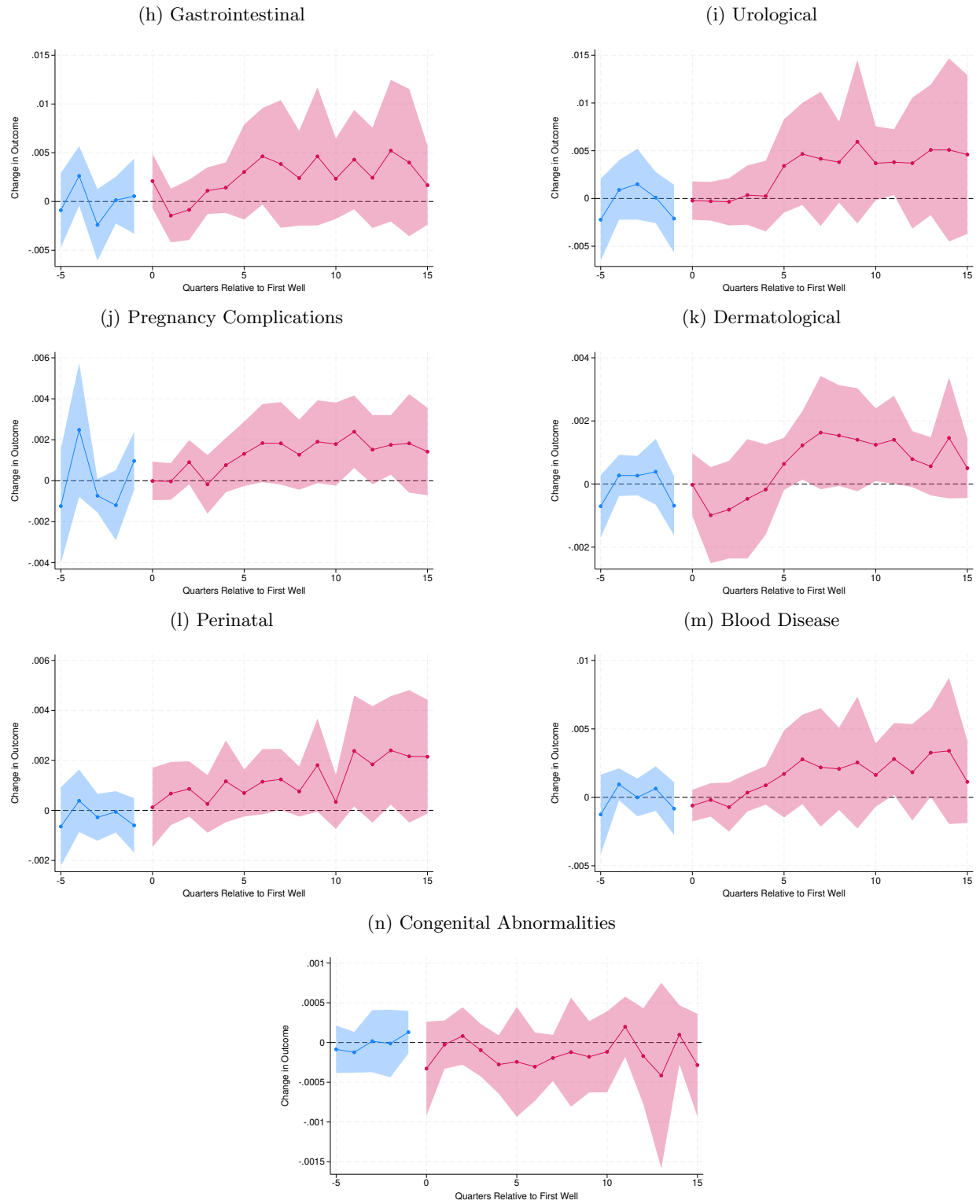
Notes: The figure above plots the coefficients and 95 percent confidence intervals for the event study corresponding to the primary analysis.

Figure A2: Condition-level healthcare utilization event studies: ZCTA-level



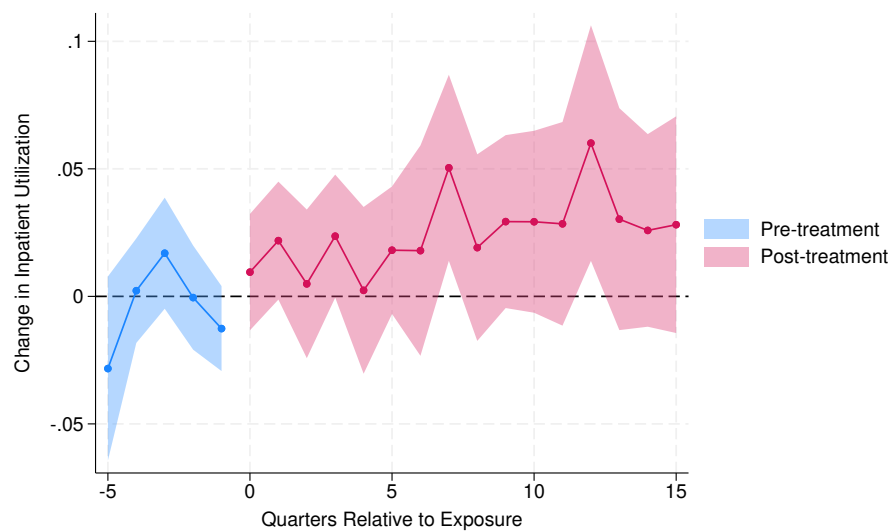
Notes: The figure above plots the coefficients and 95 percent confidence intervals for the event study corresponding to the ZCTA-level analysis for each individual condition.

Condition-level healthcare utilization event studies; continued



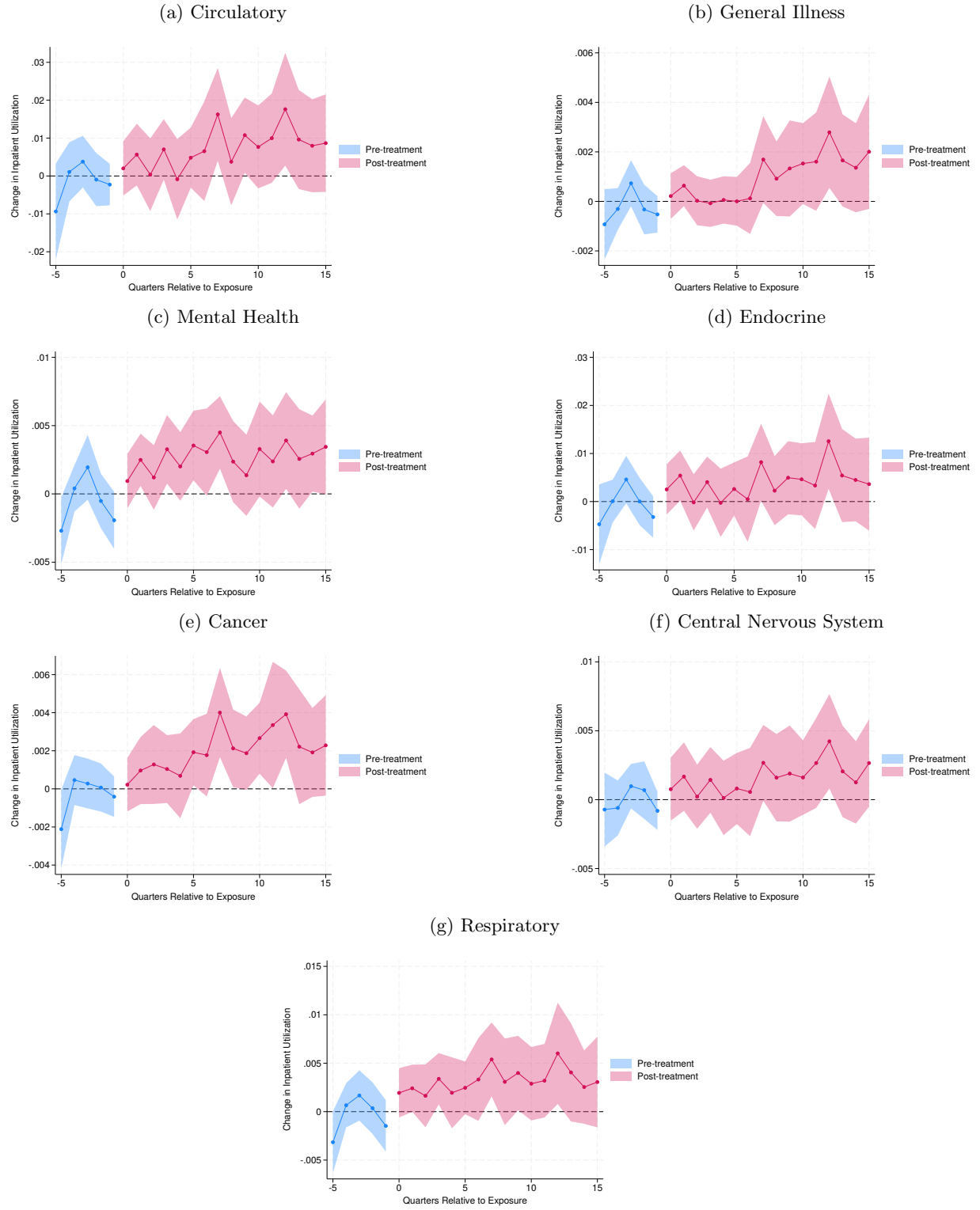
Notes: The figure above plots the coefficients and 95 percent confidence intervals for the event study that corresponds to the ZCTA-level analysis for each individual condition.

Figure A3: Fracking-related healthcare utilization event study: groundwater exposure



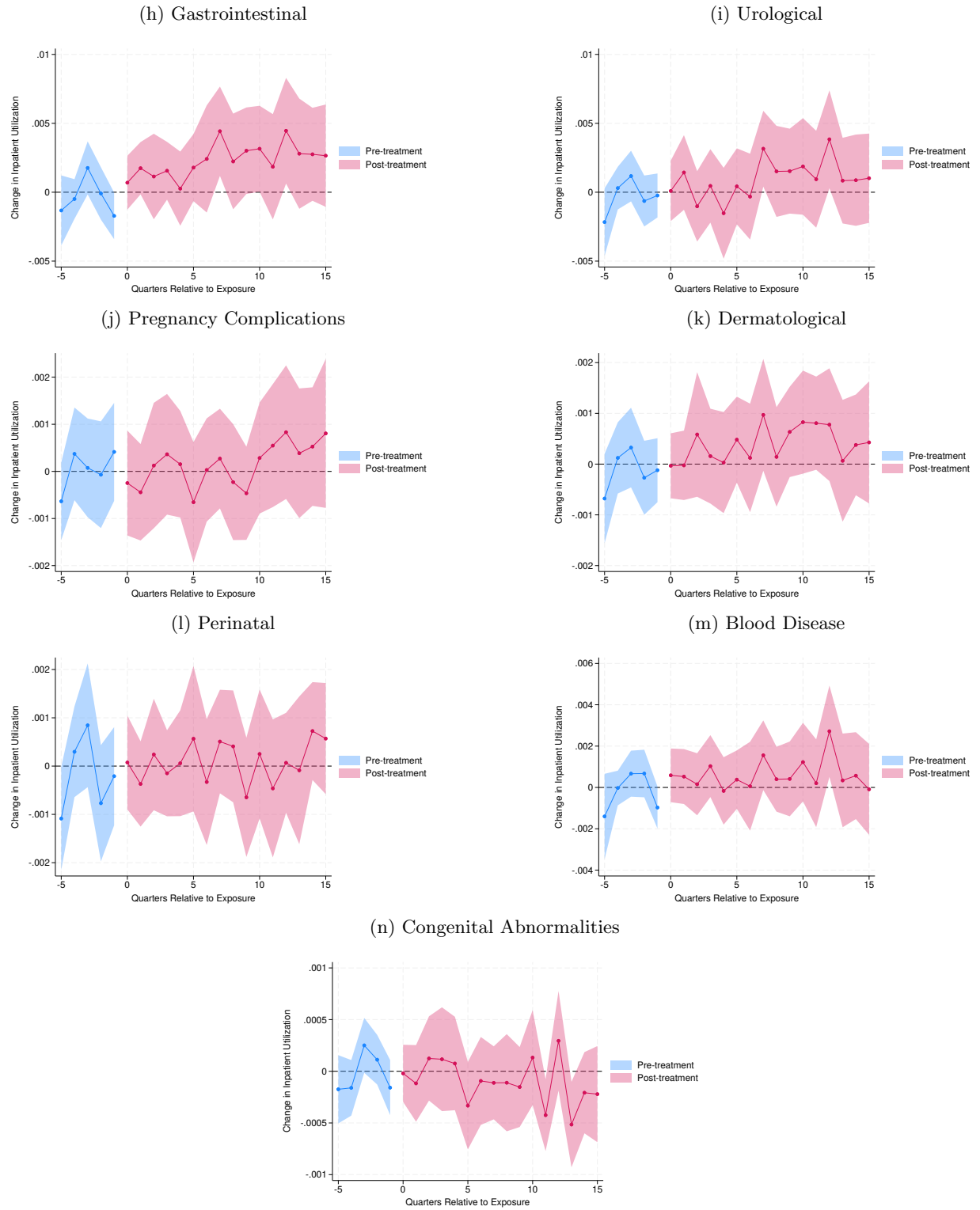
Notes: The figure above plots the coefficients and 95 percent confidence intervals for the event study corresponding to the groundwater analysis for all conditions. Treatment occurs when qualifying drilling first occurs within five kilometers of a public groundwater source serving the ZCTA.

Figure A4: Condition-level healthcare utilization event studies: groundwater exposure



Notes: The figure above plots the coefficients and 95 percent confidence intervals for the event studies corresponding to the five-kilometer groundwater analysis for each individual condition. Treatment occurs when qualifying drilling first occurs within five kilometers of a public groundwater source serving the ZCTA.

Condition-level healthcare utilization event studies: groundwater exposure; continued



Notes: The figure above plots the coefficients and 95 percent confidence intervals for the event studies corresponding to the five-kilometer groundwater analysis for each individual condition. Treatment occurs when qualifying drilling first occurs within five kilometers of a public groundwater source serving the ZCTA.